NSF PROPOSAL

Generative Problem Solving: A Constraint-Aware Foundation for Trustworthy AI in Combinatorial Optimization

[PI Name]

[Department, Institution]

[Email] | [Institution Address]

Submitted to: NSF CISE Core Programs — Algorithmic Foundations (AF) / Robust Intelligence (RI)

Bracketed fields (PI name, institution, dates) must be completed before submission via Research.gov. The Project Description below is written to fill the NSF 15-page limit at 11pt Times New Roman with standard margins. Administrative sections requiring PI-specific input are listed at the end.

Project Summary

(This Project Summary is a separate 1-page NSF component and is not counted within the 15-page Project Description limit that follows.)

Overview

Large language models (LLMs) excel at explaining optimization problems in natural language but remain unreliable when asked to solve combinatorial optimization problems that require correctness guarantees. Conversely, classical optimization algorithms — Dijkstra's algorithm, integer linear programming (ILP), dynamic programming, SAT/CSP solvers — are precise but narrowly specialized, requiring substantial expert effort to adapt to new problem domains or constraint combinations. This proposal introduces Generative Problem Solving (GPS), a research paradigm in which an AI system generates, interprets, categorizes, and solves constrained optimization problems by automatically composing verified solver pipelines rather than acting as a black-box predictor.

Preliminary work by the PI, GPS-SP (Generative Problem Solver for the Shortest Path Problem), demonstrates the core idea in a single domain: automatically generated benchmark problems, constraint-aware solver selection, and verifiable solutions for shortest-path optimization. This proposal generalizes that result into a formal theory of constraint categorization and automatic solver synthesis applicable across combinatorial optimization broadly, with a five-year research plan spanning theory, algorithms, systems, and deployment.

Intellectual Merit

The project develops a formal ontology of optimization constraints; algorithms for automatic solver composition from natural-language problem specifications; a scalable, community-usable benchmark-generation framework; correctness-verification and proof-generation methods for AI-assisted optimization; and empirical validation across shortest-path, vehicle routing, scheduling, and multi-agent planning domains. The central scientific question is whether optimization problems can be organized into computational categories that permit automatic, verifiable solver synthesis.

Broader Impacts

The resulting open-source GPS platform, benchmark datasets, and verification tools will be freely released to the research community, lowering the barrier to applying rigorous optimization methods in robotics, autonomous vehicles, smart manufacturing, supply-chain logistics, healthcare logistics, and disaster response. The project will support graduate and undergraduate research training, produce educational materials for algorithms and AI courses, and advance trustworthy-AI methodology by making optimization systems more explainable and verifiable.

Project Description

1. Introduction and Motivation

Modern AI systems, particularly large language models, are increasingly used to interpret and reason about problems posed in natural language. Users routinely describe optimization needs conversationally — "find me the cheapest route that avoids tolls and stops at a charging station" — and expect a system to both understand the request and produce a correct answer. However, when such problems involve hard combinatorial constraints — routing under budget limits, scheduling under resource contention, planning under safety constraints — LLMs alone cannot guarantee correctness. They can hallucinate feasible-looking but invalid solutions, and they have no mechanism for verifying that a proposed answer actually satisfies the stated constraints. Purpose-built optimization algorithms, on the other hand, are precise and well-understood but cannot be invoked automatically: translating a natural-language specification into a solvable formal model, and choosing the right algorithm for that model, is still a manual, expert-driven process.

This gap between natural-language flexibility and algorithmic correctness motivates the central research question of this proposal: can optimization problems themselves be organized into computational categories that permit automatic, verifiable solver synthesis? Rather than producing another specialized shortest-path algorithm, this project develops a general theory of constraint-aware optimization — Generative Problem Solving (GPS) — in which the PI's prior GPS-SP system becomes the first validation of a broader research program rather than the program's end point. The proposal reframes the role of LLMs in optimization: not as end-to-end solvers, but as problem generators, constraint interpreters, and planners that invoke, compose, and verify formal solvers.

The timing for this work is favorable. Combinatorial optimization has decades of mature algorithmic results, LLMs have recently demonstrated strong capabilities in structured natural-language understanding and code generation, and there is growing demand across robotics, logistics, and autonomous systems for optimization tools that are both flexible to specify and provably correct. GPS is designed to sit at this intersection.

1.1 A motivating scenario

Consider a disaster-response coordinator who needs to route a supply vehicle through a partially flooded city, avoiding roads known to be impassable, stopping at two designated distribution points, and returning to base within a fixed fuel budget. Today, satisfying this request requires either a human operator translating the request into a formal shortest-path-with-constraints model by hand, or an LLM that can describe a plausible-sounding route without any guarantee that the route actually avoids the flooded roads or respects the fuel budget. Neither option is well suited to a time-critical setting where both flexibility of specification and correctness of the answer matter. GPS is designed to close exactly this gap: the coordinator's natural-language request is automatically parsed into formal constraints, categorized, routed to a composed and verified solver, and returned with a human-readable explanation of why the proposed route satisfies each stated requirement. Analogous scenarios recur across the application domains discussed in Section 7.1 — a scheduling coordinator specifying resource and precedence constraints conversationally, or a fleet manager specifying capacity and routing constraints — motivating a domain-general rather than single-application solution.

2. Background and Related Work

This project draws on and extends four related lines of research.

2.1 Classical combinatorial optimization and algorithm design

Decades of algorithms research provide the solver primitives GPS composes: shortest-path search (Dijkstra, A*, Bellman-Ford), integer linear programming, dynamic programming, and SAT/CSP methods. Each is well understood in isolation, with known complexity bounds and correctness guarantees. GPS's contribution is not a new base algorithm but a principled way to select and combine these existing, verified components automatically, based on the constraint structure of a given problem instance rather than a human's manual judgment.

2.2 Constraint programming and constraint satisfaction

Constraint programming (CP) and constraint satisfaction problem (CSP) research have long studied formal constraint representations and general-purpose solvers capable of handling heterogeneous constraint types within a single modeling language. This tradition supplies useful representational ideas for the Aim 1 taxonomy. However, CP systems typically require constraints to already be specified formally in a modeling language, and do not address automatic translation from natural language, nor automatic selection among solver back-ends drawn from outside the CP family (e.g., graph-search algorithms are not naturally expressed in most CP frameworks). GPS treats CP solvers as one category of composable component rather than as the sole solving mechanism.

2.3 Neurosymbolic AI and tool-augmented language models

Recent work has shown that language models paired with external symbolic solvers or calculators can improve correctness on structured reasoning tasks relative to unaided generation. This project extends that idea substantially: rather than a single LLM-to-single-tool invocation, GPS performs full pipeline synthesis, in which multiple solver components are automatically composed based on a categorized decomposition of the original constraints. The LLM's role is also broader than in typical tool-augmented setups — it participates in constraint parsing, category identification, and (Aim 5) explanation generation, not solely in deciding when to call a tool.

2.4 Program synthesis and automatic algorithm selection

Program synthesis research addresses automatically generating code that satisfies a specification, and algorithm-selection research addresses choosing the best-performing algorithm for a given problem instance from a fixed candidate set. Both inform Aim 2's approach, but prior algorithm-selection work generally assumes a fixed, hand-built menu of whole algorithms to choose among, rather than synthesizing new pipelines by composing sub-solvers aligned to individually categorized constraints within a single problem instance.

2.5 Positioning of this proposal

What distinguishes GPS from each of these lines individually is the combination: a constraint taxonomy general enough to span multiple optimization domains (Aim 1), automatic translation from natural language into that taxonomy (Aim 2), automatic composition — not just selection — of solvers (Aim 2), a large-scale generative benchmark suite spanning domains (Aim 3), empirical validation across domains with structurally different constraint types (Aim 4), and verification of the resulting solutions (Aim 5) — integrated into a single pipeline. No existing system, to the PI's knowledge, addresses all of these simultaneously; most systems in the tool-augmented LLM literature invoke a single external solver per query rather than synthesizing a solver pipeline from categorized sub-constraints. Table 1 summarizes this positioning.

Approach	Auto NL→constraints	Multi-domain taxonomy	Composes (not just selects) solvers	Verified output
Constraint programming (CP/CSP)	No	Single modeling language	No — single solver family	Partial (solver-internal)

Approach	Auto NL→constraints	Multi-domain taxonomy	Composes (not just selects) solvers	Verified output
Tool-augmented / neurosymbolic LLMs	Yes (ad hoc)	No	No — single tool per query	Rare
Algorithm selection	No	No — fixed candidate set	No — selects, doesn't compose	No
GPS (this proposal)	Yes (Aim 2)	Yes (Aim 1)	Yes (Aim 2)	Yes (Aim 5)

This positioning also clarifies what this proposal is not claiming. It is not claiming that any individual solver component (shortest-path search, ILP, SAT) is itself novel — those are mature, well-studied algorithms and the project treats them as building blocks. It is also not claiming that natural-language-to-formal-specification translation is solved in general; Aim 2's constraint parser is scoped to the categories defined by Aim 1's taxonomy, and the Aim 5 uncertainty-estimation component is specifically designed to handle, rather than hide, the cases where that translation is unreliable. The claimed contribution is the combination described above, evaluated empirically across domains, together with the formal taxonomy that makes automatic composition possible in the first place.

3. Preliminary Work: GPS-SP

The PI's prior work, GPS-SP (Generative Problem Solver for the Shortest Path Problem), established feasibility for the core GPS idea within a single, well-understood domain. GPS-SP combines automatic generation of shortest-path benchmark problems with constraint-aware solver selection, producing solutions more efficiently and more reliably than exhaustive search or general-purpose AI baselines. Three aspects of GPS-SP directly seed the aims below.

3.1 Constraint-aware solver selection

GPS-SP demonstrated that even within a single problem class (shortest path), the presence of different constraint types — edge weights only, forbidden nodes, required waypoints, budget limits — favors different solution strategies, and that a system can be trained to recognize which strategy to apply. This is the informal precursor to the Constraint Taxonomy formalized in Aim 1.

3.2 Generative benchmarking

GPS-SP's benchmark problems were generated automatically with controllable difficulty and automatically labeled with ground-truth solutions, avoiding the bottleneck of manually curated optimization benchmarks. This generation methodology is the seed for the large-scale, multi-domain benchmark suite proposed in Aim 3.

3.3 Efficiency and correctness gains

Relative to exhaustive search and general-purpose AI baselines, GPS-SP showed measurable improvements in both solution quality and computational efficiency on shortest-path benchmarks. This preliminary result is the evidentiary basis for the proposal's central hypothesis: that constraint categorization can meaningfully guide automatic solver choice, and that automatically generated, labeled benchmarks can support both training and rigorous evaluation at scale.

3.4 Limitations of GPS-SP motivating this proposal

GPS-SP's scope is intentionally narrow, and three specific limitations directly motivate the aims below. First, its constraint categorization is informal and hand-built for shortest-path problems specifically; it is not clear a priori whether the same categories, or a compatible generalization of them, apply to problems with fundamentally different structure such as scheduling or multi-agent coordination — this motivates Aim 1's formalization and Aim 4's cross-domain test. Second, GPS-SP selects among a small, fixed menu

of shortest-path solution strategies; it does not compose solver components together, and has no mechanism for handling a domain it was not built for — this motivates Aim 2's composition-based approach. Third, GPS-SP's outputs are evaluated against ground truth during benchmarking but are not accompanied by a runtime, user-facing correctness certificate — this motivates Aim 5's verification and explainability layer. The remainder of this Project Description describes how the five proposed aims address each of these limitations and generalize GPS-SP's three components into a formal, domain-independent framework. Table references throughout (Tables 1–3) point to the comparison, benchmark-scale, and risk summaries introduced in later sections, so that the aims below can be read either sequentially or by consulting a specific aim in isolation.

4. Research Objectives

The project is organized into five aims, executed largely in sequence but with substantial overlap, as detailed in the timeline (Section 12). Aims 1–3 build the theoretical and infrastructural foundation; Aim 4 validates that foundation across domains; Aim 5 addresses the reliability requirements needed for real-world deployment. The aims are deliberately structured with feedback loops rather than as a strict waterfall: Aim 4's per-domain evaluation results are expected to surface cases where the Aim 1 taxonomy needs revision, and those revisions in turn affect Aim 2's composition engine, so the team plans for iterative refinement across Years 2–4 rather than treating Aim 1 as fixed after Year 1. This structure is reflected in the annual internal-review process described in Section 9 and the evaluation criteria in Section 11.

Aim 1 — Formal Ontology of Optimization Constraints

This aim develops a constraint taxonomy general enough to span shortest-path, routing, scheduling, and resource-allocation problems.

Approach

Starting from GPS-SP's informal constraint categories, the PI and team will (1) survey constraint types across representative combinatorial optimization domains, (2) identify structural equivalence classes — constraints that, despite surface differences, admit the same solution strategy (e.g., "forbidden node" in routing and "forbidden time slot" in scheduling are structurally similar exclusion constraints), (3) define a composition operator allowing multiple constraint categories to be combined for problems with interacting constraints, and (4) formalize the taxonomy as a machine-readable schema that downstream modules (Aim 2) can query.

Key questions

- What constraint categories are universal across optimization domains, versus domain-specific?
- Can arbitrary natural-language constraints be mapped automatically into this taxonomy, and with what accuracy?
- Can categories be composed to describe problems with multiple interacting constraints without combinatorial blow-up in the taxonomy itself?

Toward a formal definition

Informally, the taxonomy defines a constraint category as an equivalence class of constraints that admit a common, parameterized solution strategy. A category is specified by (i) a schema describing the natural-language and formal-logic patterns that instantiate it, (ii) the set of solver components capable of handling it, and (iii) composition rules describing how it interacts with other categories when present in the same problem (e.g., an exclusion constraint composed with a budget constraint prunes the same search space differently than either alone). Early candidate categories drawn from GPS-SP and the Aim 4 domains include exclusion (forbidden nodes/times/resources), inclusion/waypoint (required visits or assignments), capacity (bounded resource consumption), precedence (ordering requirements), and coordination (multi-agent interaction constraints). This candidate list is expected to evolve during Year 1 as it is tested against real benchmark instances.

Evaluation and deliverable

The taxonomy will be evaluated by (a) coverage — the fraction of constraints in the Aim 3 benchmark suite that map cleanly into a taxonomy category, and (b) inter-annotator agreement when domain experts independently categorize a held-out sample of constraints. The deliverable is Constraint Taxonomy 2.0, a versioned, published specification extending GPS-SP's informal categorization.

Aim 2 — Automatic Solver Composition

Rather than requiring a human to select among Dijkstra's algorithm, ILP, dynamic programming, SAT, or CSP methods, this aim develops a pipeline that performs the selection automatically.

Approach

The pipeline has three stages. A constraint parser (built on an LLM fine-tuned or prompted for structured extraction) converts a natural-language problem statement into a set of formal constraint instances. A category-identification module matches each extracted constraint against the Aim 1 taxonomy, flagging low-confidence matches for clarification rather than guessing. A solver-construction module then assembles a solver pipeline from a library of verified solver components, executing it and returning a solution together with the constraint mapping that justifies it. For example, the query "find the shortest route avoiding hospitals, visiting two charging stations, and remaining under a fuel budget" is parsed into an exclusion constraint (hospitals), a waypoint constraint (charging stations), and a budget constraint, categorized against the taxonomy, and routed to a composed solver combining shortest-path search with waypoint ordering and budget-feasibility checking.

Technical challenges

- Ensuring the constraint parser's output is well-formed and complete relative to the original natural-language statement (no dropped or invented constraints).
- Composing solver components that were not originally designed to interoperate, without sacrificing correctness.
- Handling cases where no known composition satisfies all extracted constraints, and communicating infeasibility clearly rather than returning a degraded answer.

Worked example

Consider again the query "find the shortest route avoiding hospitals, visiting two charging stations, and remaining under a fuel budget." The constraint parser first extracts three formal constraint instances: an exclusion set (hospital locations), an ordered inclusion requirement (two charging-station visits, order unspecified), and a numeric budget bound (fuel). The category-identification module matches these against Aim 1's taxonomy: exclusion, waypoint/inclusion, and capacity, respectively, each with an associated confidence score. Because all three scores exceed the acceptance threshold, the solver-construction module proceeds directly to composition, combining a constrained shortest-path search (handling exclusion) with a waypoint-ordering step (handling inclusion) and a budget-feasibility filter (handling capacity) into a single executable pipeline. If, instead, the parser had produced a low-confidence match — for example, an ambiguous phrase like "avoid busy areas" with no clear formal mapping — the pipeline would surface a clarification request rather than silently guessing a definition of "busy," consistent with the Aim 5 uncertainty-estimation design.

Evaluation and deliverable

Composed solvers will be benchmarked against hand-designed, domain-specific baselines on solution quality, runtime, and constraint-satisfaction rate. The deliverable is an open-source solver-composition engine and an accompanying technical report characterizing where automatic composition matches, exceeds, or falls short of hand-tuned baselines.

Aim 3 — Generative Benchmark Creation

This aim extends the benchmark-generation approach demonstrated in GPS-SP from a single domain to a large, multi-domain, difficulty-controllable benchmark suite.

Approach

Benchmark instances will be generated programmatically with parameterized difficulty (problem size, constraint density, constraint interaction complexity) and automatically labeled with verified ground-truth solutions using independent solvers, so that correctness of generated labels does not depend on the GPS system under evaluation. Generation will proceed domain by domain, aligned with Aim 4's rollout schedule, so that each new domain added to GPS has a matching benchmark before evaluation begins.

Community resource design

The benchmark suite will be released under an open license with versioned releases, a standardized problem-instance format shared across domains (to support cross-domain tooling), and a public leaderboard to encourage adoption by the broader optimization and AI research community, extending the resource's value beyond this project's internal evaluation needs.

Target scale by domain

Table 2 gives preliminary per-domain instance-count targets, aligned to the Aim 4 rollout schedule; these targets will be revisited annually based on Aim 4 evaluation needs and computational cost.

Domain	Year introduced	Target instances by Y5	Difficulty tiers
Shortest path (from GPS-SP)	Y1 (refined)	≥ 500,000	3 (easy/medium/hard)
Minimum spanning tree	Y2	≥ 300,000	3
Vehicle routing	Y3	≥ 400,000	4 (adds capacity tiers)
Scheduling	Y3	≥ 400,000	4 (adds precedence tiers)
Resource allocation	Y3	≥ 300,000	3
Multi-agent planning	Y4	≥ 250,000	4 (adds coordination tiers)

Reproducibility and open science

Every released benchmark version will be accompanied by a fixed random seed and generation-code version, so that results reported against a given benchmark version are exactly reproducible by other researchers. Ground-truth labels will be produced by solvers independent of the GPS composition engine wherever possible, so that benchmark correctness does not depend on the system being evaluated against it — an important methodological safeguard against inadvertently building a benchmark that only validates GPS's own design choices.

Evaluation and deliverable

Benchmark quality will be assessed via difficulty-distribution analysis, diversity metrics relative to hand-curated benchmarks in the literature, and periodic expert audits of generated instances. The deliverable is a versioned, public benchmark suite (targeting on the order of millions of labeled instances across domains by Year 5, per Table 2) with accompanying generation code.

Aim 4 — Generalization Beyond Shortest Path

The GPS framework developed in Aims 1–3 will be demonstrated on the Traveling Salesman Problem, Vehicle Routing, Minimum Spanning Tree, scheduling, resource allocation, and multi-agent planning.

Domain selection rationale

These domains were chosen because they span constraint structures largely absent from shortest-path problems: sequencing constraints (TSP, scheduling), capacity constraints (vehicle routing, resource allocation), precedence constraints (scheduling), and coordination constraints (multi-agent planning).

Success across this set is a stringent test of Aim 1's claim that the taxonomy is domain-independent rather than an artifact of the shortest-path case.

Approach

For each domain, the team will (1) identify existing hand-designed algorithms to serve as baselines, (2) apply the Aim 2 composition pipeline using only the Aim 1 taxonomy and Aim 3 benchmarks for that domain — without domain-specific engineering of the composition engine itself — and (3) measure the gap, if any, between composed and hand-designed solvers. Persistent gaps in a given domain will feed back into Aim 1 as evidence that the taxonomy needs revision, closing the loop between theory and validation.

Per-domain notes

Traveling Salesman and Vehicle Routing introduce sequencing and capacity constraints not present in shortest-path problems; baselines include established heuristic and exact methods for these NP-hard problems, and GPS is expected to be evaluated primarily on small-to-moderate instance sizes where exact or near-exact composed solutions are tractable. Minimum Spanning Tree is included specifically as a structurally simple domain to validate that GPS does not over-fit its taxonomy to the harder cases, and provides an early, low-risk validation point in Year 3. Scheduling introduces precedence and resource-contention constraints, testing the taxonomy's inclusion category in a substantially different setting than routing waypoints. Resource allocation tests capacity constraints without an underlying graph structure, probing whether the taxonomy's categories are genuinely structural rather than tied to graph-based problems. Multi-agent planning, targeted for Year 4, introduces coordination constraints between multiple simultaneous solutions and is treated as the most demanding generalization test in the research plan, since it requires composing solvers whose outputs must additionally remain mutually consistent.

Evaluation and deliverable

Success is measured primarily by constraint-satisfaction rate and solution-quality ratio relative to domain-specific baselines, tracked per domain and in aggregate. The deliverable is a multi-domain GPS prototype with published per-domain evaluation results.

Aim 5 — Trustworthy AI: Verification and Explainability

Because optimization outputs are frequently used in safety- or cost-critical settings, this aim develops mechanisms for establishing and communicating trust in GPS-produced solutions.

Approach

Three components are developed in parallel. Correctness certificates: for a given composed solver's output, generate a machine-checkable certificate that the solution satisfies each originally extracted constraint, independent of whether the composition itself was correct — i.e., verifying the answer rather than the derivation. Explainability: produce a human-readable justification tracing each constraint in the original natural-language request to the specific part of the solution that satisfies it. Uncertainty estimation: quantify confidence in the constraint-parsing stage (Aim 2) so that ambiguous natural-language requests are flagged for user clarification rather than silently resolved by a guess.

Evaluation and deliverable

Certificates will be evaluated for soundness (do they ever pass an incorrect solution?) and coverage (what fraction of composed-solver outputs can be certified within a practical time budget). Explainability outputs will be evaluated with human-subject studies measuring whether domain-expert users can correctly predict, from the explanation, whether a solution is valid. The deliverable is a verification and explainability layer integrated into the GPS platform, plus a published methodology for certificate generation in AI-assisted combinatorial optimization.

Human-subject evaluation design

The explainability evaluation will follow a structured protocol: domain-expert participants (recruited from the PI's institution and, where available, external collaborators in the Aim 4 application domains) will be shown a GPS-generated solution along with its explanation and asked to judge whether the solution satisfies each stated constraint, without independently re-solving the problem from scratch. Accuracy and response time on this judgment task, compared against a baseline condition with no explanation provided, will serve as the primary explainability metric. This protocol will be submitted for institutional human-subjects review (IRB) prior to Year 3 data collection, consistent with standard practice for studies involving human participants.

5. Research Questions

- Can optimization constraints be classified independently of application domain?
- Can automatic solver synthesis match or outperform hand-designed, domain-specific algorithms, and under what conditions does it fall short?
- Can LLMs function reliably as planners and constraint interpreters rather than as black-box optimizers?
- Can correctness certificates for optimization solutions be generated automatically, and how does certificate coverage scale with problem size?
- Can the resulting optimization pipeline remain explainable to a non-expert user without sacrificing solution quality?

These five questions are addressed respectively, though not exclusively, by Aims 1, 4, 2, 5, and 5, and are revisited explicitly in the Year 5 summary publication (Section 12) so that the project's answer to each — including any negative or partial results — is reported rather than only the aims that succeeded as hypothesized. Negative results are treated as scientifically meaningful here: for example, a domain in which automatic solver synthesis cannot match a hand-designed baseline (Research Question 2) would itself be an informative boundary condition on the GPS paradigm, and would be reported as such rather than omitted.

6. Innovation

The proposal's novelty spans three dimensions. Scientifically, it introduces a computational theory of constraint categorization that is domain-independent, in contrast to the domain-specific formulations typical of the optimization literature, and that is validated empirically across structurally distinct domains rather than proposed purely theoretically. Algorithmically, it introduces automatic synthesis — not merely selection — of optimization solvers from categorized constraints, composing existing verified components into new pipelines rather than choosing a single algorithm from a fixed menu. From an AI-systems perspective, it repositions LLMs within the optimization pipeline: as problem generators, constraint interpreters, and planners that invoke and verify formal solvers, rather than as end-to-end black-box solvers whose outputs cannot be certified. Taken together, these contributions aim to establish generative problem solving as a recognized research area distinct from, but building on, constraint programming, algorithm selection, and tool-augmented LLM reasoning.

A secondary but related innovation is methodological: the project's evaluation design (Section 11) treats cross-domain generalization gaps as first-class research findings rather than implementation bugs to be hidden, which the PI expects to produce a more honest and more useful account of where automatic, taxonomy-driven solver synthesis works well and where hand-designed, domain-specific algorithms remain preferable — a distinction that matters for anyone deciding whether to adopt a GPS-style system for a new optimization problem in practice.

7. Broader Impacts

7.1 Application domains

The GPS framework targets application domains where reliable, explainable optimization has direct societal value. In robotics and autonomous vehicles, routing and task-allocation decisions must be both fast and verifiably correct, particularly when operating near people; the Aim 5 certificate mechanism gives such systems a checkable argument rather than an opaque recommendation before a decision is acted on. In smart manufacturing, scheduling errors carry real production costs, and the scheduling-domain results from Aim 4 are directly relevant to production-line and job-shop scheduling problems. In space-mission planning, resource constraints (power, time, communication windows) are unforgiving and errors are costly to correct after the fact, making pre-execution verification especially valuable. In supply-chain and healthcare

logistics, delivery routing directly affects patient and public outcomes, and the vehicle-routing and resource-allocation results from Aim 4 map naturally onto these settings. In disaster response — the scenario introduced in Section 1.1 — rapid, trustworthy routing and resource-allocation decisions can be life-critical, and GPS's combination of natural-language flexibility and verifiable output is intended to support exactly this kind of time-pressured decision-making.

7.2 Open resources

The project will release, under open licenses, the GPS software platform, the Aim 3 multi-domain benchmark suite, and the Aim 1 constraint taxonomy specification. Open release is intended to let other research groups build on the taxonomy and benchmarks without reproducing GPS-SP-scale preliminary work themselves, lowering the barrier to entry for constraint-aware AI research broadly. The PI anticipates that the taxonomy specification in particular has value independent of the GPS software itself: a research group building an unrelated optimization system could adopt the taxonomy as a shared vocabulary for describing constraint types, similar to how shared benchmark formats have lowered coordination costs in other areas of machine learning research.

7.3 Education and workforce development

The project will produce educational modules suitable for undergraduate and graduate coursework in algorithms, AI, and optimization, built around the Aim 3 benchmark suite so that students can experiment with real, verifiable problem instances rather than toy examples. These modules are planned as drop-in units (roughly 1–2 weeks of course content each) covering constraint categorization, solver composition, and verification, suitable for adoption in existing algorithms or AI courses at the PI's and collaborating institutions rather than requiring a new standalone course. The five-year plan is structured to support multiple graduate students across its duration — with responsibilities distributed across the five aims — and to create structured undergraduate research opportunities (e.g., benchmark curation, domain-specific solver integration in Aim 4, evaluation-harness development) suitable for students early in a research career and scoped to a single semester or summer term.

7.4 Broadening participation

The PI will recruit students through existing institutional programs supporting participation of students from groups underrepresented in computing, and will structure Aim 3 and Aim 4 tasks (benchmark curation, domain extension) to provide well-scoped entry points for early-career researchers, including undergraduates, without requiring prior expertise in formal optimization theory. Where the PI's institution supports summer research programs (e.g., REU-style opportunities) or K-12 outreach events, the project will offer a lightweight, visual demonstration of GPS (natural-language problem in, verified solution and explanation out) suitable for those settings.

7.5 Dissemination

Results will be disseminated through peer-reviewed publication at relevant AI and algorithms venues, open-source releases with documentation, and presentation at NSF-relevant PI meetings. Where domain partners exist (e.g., robotics or logistics collaborators), the project will pursue applied case studies to support technology transition in Year 5. All software and benchmark releases will include versioned documentation and example notebooks to reduce the effort required for external researchers and educators to adopt the platform.

8. Ethical Considerations

GPS is designed to produce verifiable answers rather than merely persuasive-sounding ones, which the PI views as a direct ethical benefit relative to unaided LLM use in decision-relevant settings: a user who receives a GPS output also receives a checkable argument for why it satisfies their stated constraints, rather

than a confident-sounding but unverified claim. That said, two considerations warrant explicit attention throughout the project. First, correctness certificates (Aim 5) verify that a solution satisfies the constraints as parsed — they do not, and cannot, guarantee that the parser correctly captured the user's intent; the uncertainty-estimation component is therefore treated as a safety-relevant feature, not an optional one, and low-confidence parses will default to requesting clarification rather than proceeding. Second, because target application domains include safety-relevant settings such as disaster response and autonomous vehicles, the project will clearly document GPS's validated operating conditions (problem domains and scales evaluated under Aim 4) and will not represent the system as suitable for domains or scales beyond what has been empirically validated.

9. Management Plan

The PI will have overall responsibility for scientific direction and NSF reporting. Graduate students will be assigned primary ownership of individual aims (e.g., one student leading Aim 1/Aim 2 taxonomy-and-composition work, another leading Aim 3 benchmark infrastructure, a third leading Aim 5 verification), with the PI coordinating integration across aims at regular project meetings. Undergraduate researchers will be paired with graduate-student mentors on well-scoped sub-tasks within Aim 3 and Aim 4, as described in Section 7.3–7.4.

Progress will be tracked against the year-by-year milestones in Section 12 and the quantitative criteria in Section 11, with internal review at the end of each project year to decide whether a given aim proceeds as planned, is revised (e.g., taxonomy refinement triggered by Aim 4 results), or is re-scoped. Where domain expertise beyond the core team is needed — for example, in scheduling or multi-agent planning for Aim 4 — the PI will seek informal consultation with colleagues in those areas and, where appropriate, formal collaboration.

The PI's prior work on GPS-SP directly demonstrates the technical background needed to lead this project: design and evaluation of constraint-aware solver-selection methods, generative benchmark construction, and empirical comparison against exhaustive-search and general-purpose-AI baselines. This preliminary experience is the basis for the Year 1 milestone of refining GPS-SP into the taxonomy-driven framework described in Aim 1, rather than starting the formal theory from a blank slate.

10. Risk Assessment and Mitigation

As with any multi-year systems and theory project, several risks could affect the pace or scope of the research plan. Table 2 summarizes the primary risks identified for this proposal and the corresponding mitigation strategy built into the research plan.

Risk	Likelihood	Mitigation
Constraint taxonomy fails to generalize cleanly across domains	Medium	Begin with a small set of well-understood domains (Aim 4) chosen for structurally distinct constraint types; iteratively refine the taxonomy rather than fixing it in Year 1
Automatic solver synthesis underperforms hand-tuned domain algorithms	Medium	Treat hand-designed algorithms as baselines throughout, not just at the end; allow the composition engine to fall back to known domain-specific solvers when synthesis confidence is low
Natural-language constraint parsing introduces ambiguity or error	High	Pair parsing with the Aim 5 uncertainty-estimation module; flag low-confidence parses for user clarification rather than silently proceeding

Risk	Likelihood	Mitigation
Benchmark generation at scale produces low-diversity or trivial problems	Low–Medium	Use difficulty-controllable generation methods validated in GPS-SP; periodically audit samples with domain experts
Verification/proof generation does not scale to large problem instances	Medium	Prioritize certificate generation for solution correctness on the final composed solver's output rather than full formal verification of the synthesis process itself

These risks are treated as design inputs rather than after-the-fact caveats. For instance, the decision to prioritize solution-level correctness certificates over full formal verification of the synthesis process itself (Aim 5) is a direct response to the scalability risk identified above, and the decision to sequence Aim 4 domains from structurally simple (Minimum Spanning Tree) to structurally demanding (multi-agent planning) is a direct response to the taxonomy-generalization risk. The PI will revisit this risk register at each annual internal review (Section 9) and update mitigation strategies as empirical results from Aim 4 accumulate.

11. Evaluation Plan

Progress will be assessed against explicit, quantitative criteria at each project year, reviewed internally by the PI and team and reported in annual NSF progress reports. Where a criterion is not met, the internal review process (Section 9) determines whether the corresponding aim should be revised, given additional time, or re-scoped, rather than treating a missed target as a project-ending failure; the risk-mitigation strategies in Section 10 are designed with this iterative adjustment in mind.

- Aim 1: taxonomy coverage ≥ target threshold on held-out Aim 3 benchmark constraints; inter-annotator agreement above an established acceptability threshold.
- Aim 2: composed-solver constraint-satisfaction rate and solution-quality ratio relative to hand-designed baselines, tracked per domain as Aim 4 domains come online.
- Aim 3: benchmark suite size, diversity, and difficulty-distribution metrics, with periodic expert audits.
- Aim 4: per-domain gap between composed and hand-designed solvers, used as a direct signal for taxonomy refinement (feedback into Aim 1).
- Aim 5: certificate soundness and coverage; human-subject explainability evaluation results.

This evaluation structure creates explicit feedback loops between the aims — most importantly, Aim 4's cross-domain results are treated as ongoing evidence for or against Aim 1's central taxonomy claim, rather than a one-time final validation step, so that the taxonomy can be revised iteratively rather than only at the end of the project.

12. Five-Year Research Plan

The plan below sequences the five aims across five years, with Aims 1–3 front-loaded to establish the shared foundation used by Aims 4 and 5.

Year	Focus	Key Milestones	Deliverables
1	Formal theory & preliminary refinement	Formalize Constraint Taxonomy 2.0; extend GPS-SP's implicit categorization into a machine-usable schema; recruit and train first graduate students; establish evaluation infrastructure	Taxonomy specification document; refined GPS-SP implementation; workshop paper
2	Solver synthesis & benchmark infrastructure	Implement constraint-parsing and category-identification modules (Aim 2); build automatic solver-composition engine; begin large-scale benchmark generation (Aim 3); design verification framework (Aim 5)	GPS solver-composition prototype (v1); benchmark generator (alpha); first conference submission
3	Cross-domain validation I	Extend GPS to scheduling, vehicle routing, and resource allocation (Aim 4); expand benchmark suite to these domains; integrate initial correctness-certificate generation	Multi-domain GPS prototype (v2); public benchmark release (v1); 1–2 peer-reviewed publications
4	Cross-domain validation II & verification	Extend to robotics, multi-agent planning, and scientific-workflow scheduling; mature proof-generation and explainability components (Aim 5); user studies with domain-expert collaborators	GPS platform (v3) with verification layer; usability study results; educational materials draft
5	Platform release & dissemination	Finalize general GPS platform architecture; full community benchmark release; open-source software release with documentation;	Public open-source release; final benchmark suite (v2); summary publication; course modules

Year	Focus	Key Milestones	Deliverables
		technology-transition activities with industry/robotics partners	

13. Alignment with NSF Programs

This work aligns with several NSF programs and could be tailored toward whichever best matches the target solicitation:

- NSF CISE Core Programs — Algorithmic Foundations (AF): theoretical constraint-categorization and solver-synthesis contributions (Aims 1–2).
- NSF Robust Intelligence (RI): trustworthy, verifiable AI reasoning over constraints (Aim 5).
- NSF National AI Research Institutes: as a component of a larger, multi-PI center proposal spanning theory, systems, and applications.
- NSF Foundational Research in Robotics (FRR): robotics and autonomous-systems applications (Aim 4).
- NSF Smart and Connected Communities (S&CC): urban-infrastructure and logistics optimization applications (Section 7.1).

Recommendation: submit to a single core program (AF or RI) for this proposal, and reserve the multi-domain, multi-institution version for a future Institute-scale submission, since NSF core awards are evaluated more favorably when scoped to one program's review criteria.

14. Summary: Intellectual Merit and Broader Impacts

Looking beyond the five-year period, a successful outcome of this project positions Generative Problem Solving as a foundation that later work — by this team or others — could extend toward fully automated, verified decision-support systems for domains that today rely on either manual formal modeling or unverified AI assistance. A general, taxonomy-driven, verifiable pipeline for combinatorial optimization would be a reusable building block for exactly the kind of high-stakes, natural-language-specified planning problems that motivate Section 1.1's disaster-response scenario, and the open datasets, taxonomy specification, and software released over the course of the project are intended to let that extension happen in the broader research community rather than only within this project.

Relative to the PI's published GPS-SP work, this proposal shifts the central claim from "a faster shortest-path solver" to "a general theory and architecture for Generative Problem Solving" across combinatorial optimization. GPS-SP is retained as strong preliminary evidence that constraint categorization and automatic solver selection improve efficiency and correctness relative to exhaustive search and general-purpose AI approaches, while the proposed five-year program of formal theory, automatic solver synthesis, large-scale benchmarking, cross-domain generalization, and verification methods constitutes a fundamental research contribution appropriate for NSF's evaluation criteria of intellectual merit and broader impacts.

Remaining Required Sections (PI Input Needed)

The following sections are required by NSF's Proposal & Award Policies & Procedures Guide (PAPPG) but depend on information this document does not have — please supply the details noted and they can be drafted next:

- References Cited — full citation for the GPS-SP paper and any related prior work in shortest-path, combinatorial optimization, constraint programming, and LLM-based reasoning that should be cited.
- Biographical Sketch(es) — NSF-approved format (SciENcv) for the PI and any co-PIs.
- Budget and Budget Justification — personnel, equipment, travel, and indirect costs for the 5-year period.
- Facilities, Equipment, and Other Resources — computing resources (e.g., GPU/cluster access) available at the PI's institution.
- Data Management and Sharing Plan — plan for releasing the Aim 3 benchmark suite and open-source software.
- Current and Pending (Other) Support — for the PI and any co-PIs, per NSF's current disclosure requirements.